Effects of Quercetin on 3T3-L1 Cell Lipid Accumulation at Different Adipogenic Stages

Farah Cisse[1]*, Victoria Robinson[1], Avani Ghate[1]

[1]University of California, Berkeley

Summary

This study investigates the impact of quercetin on adipogenesis and lipid accumulation in 3T3-L1 cells, with a focus on the timing of quercetin treatment during the adipogenic protocol. The motivation was inspired by a previous study done on the impacts of quercetin on liver cells in a lipogenic environment. Quercetin's anti-inflammatory and antioxidant properties make it a drug of interest for potential therapeutic implication for managing obesity-related diseases. In this experiment, quercetin was administered at three different stages of differentiation to a 3T3-L1 cell model: induction, maturation, and maintenance. Computational quantification of lipid accumulation revealed that early exposure to quercetin (during induction) significantly reduced lipid accumulation, while later exposure (during maturation and maintenance) had less inhibitory effects, even promoting lipid storage. These findings suggest that quercetin's effect on adipogenesis is timing dependent. However, more intensive exploration and long-term future studies are needed to understand the true therapeutic potential of quercetin.

1. Introduction

Obesity has become a major global concern, with the disease more than doubling since 1990, according to the World Health Organization. Spreading almost worldwide, many are at risk of developing various metabolic and cardiovascular diseases[1]. Obesity is closely linked to adipogenesis, the process by which preadipocytes (immature fat cells) differentiate into mature adipocytes (fat cells)[1]. When energy intake exceeds expenditure, the body stores the excess energy in the form of triglycerides with adipocytes. In obesity, excessive adipocyte expansion occurs leading to high lipid accumulation[2].

This study looks to explore therapeutic options for obesity in a specific compound called, quercetin. Quercetin is a flavonoid found in fruits and vegetables, containing antioxidant and anti-inflammatory properties. Quercetin exerts its effects on a molecular and cellular level through several mechanisms. It's antioxidant properties enable it to scavenge reactive oxygen

---

1

1

1

1

species (ROS) in the body[3]. These agents can damage cellular components that contribute to diseases. By suppressing ROS, quercetin helps to reduce oxidative stress in cells. Quercetin can attribute its anti-inflammatory effects to its ability to inhibit the NF-κB (Nuclear Factor kappa B) signaling pathway, which plays a central role in inflammation[3]. By inhibiting NF-κB, quercetin reduces the production of pro-inflammatory cytokines, which are often upregulated in chronic conditions like obesity[4]. Furthermore, a key driver of lipid accumulation is de novo lipogenesis (DNL), a process where excess carbohydrates are converted into fatty acids. This process is known to occur in obesity related diseases. Transcription factors such as SREBP-1 and XBP-1 are key to regulating DNL as they regulate genes involved in lipid synthesis[5]. These genes are activated under high glucose conditions with insulin resistance and endoplasmic reticulum (ER) stress. Quercetin has been shown to suppress DNL by downregulating these key transcription factors, reducing lipid synthesis[5].

The work done in this paper has drawn inspiration from research that explores Quercetin as an agent that reduces lipid accumulation in a cell model of Non-alcoholic fatty liver disease (NAFLD), a disease closely linked to lipid droplet accumulation in liver cells[5]. This research cultured HepG2 cells (human liver cell lines) in high and low glucose conditions, exposing them to quercetin to examine its effects on lipid accumulation. The study found that quercetin significantly reduced lipid accumulation in HepG2 cells, particularly under conditions of high glucose and FFAs, which typically exacerbate lipid storage in the liver[5]. Quercetin inhibited the expression of SREBP-1 and XBP-1.

This project aims to build on this body of research by exploring the effects of quercetin on adipogenesis in 3T3-L1 cells (mouse fibroblast-like cell line), a commonly used model for studying fat cell differentiation.

Keywords: Quercetin, Adipogenesis, De Novo Lipogenesis, NF-κB, Reactive Oxygen Species, Non-alcoholic Fatty Liver Disease, SREBP-1, XBP-1, Obesity, 3T3-L1 Cells

2.Material and Methods

2.1 Experimental Setup

The group maintained a 3T3-L1 mouse cell line given by the instructional team. The cell line used was passage 12. As seen in ***figure 1***, two 6-well plates were used to culture the cells. There were two control wells: The first was a negative control (well 1) that was not subject to the adipogenic protocol, meaning that the cells never received any other reagents except for culture media (CM). The second was a differentiation media (DM) only control (well 2) that was subject to the normal adipogenic protocol but never received any quercetin. The rest of the wells (wells 3-11) were experimental wells. We divided these experimental wells into three sets: The first set (wells 3-5) were the cells subject to quercetin at the earliest stage, induction of differentiation. The second set (wells 6-8) were the cells subject to quercetin at the second earliest stage, adipocyte maturation. Lastly, the third set (wells 9-11) were the cells subject to quercetin at a third time point a few days after adipocyte maturation. After each step of the protocol, the cells were incubated at 37°C until the next step. This is the optimal temperature to keep the cells in for successful differentiation.

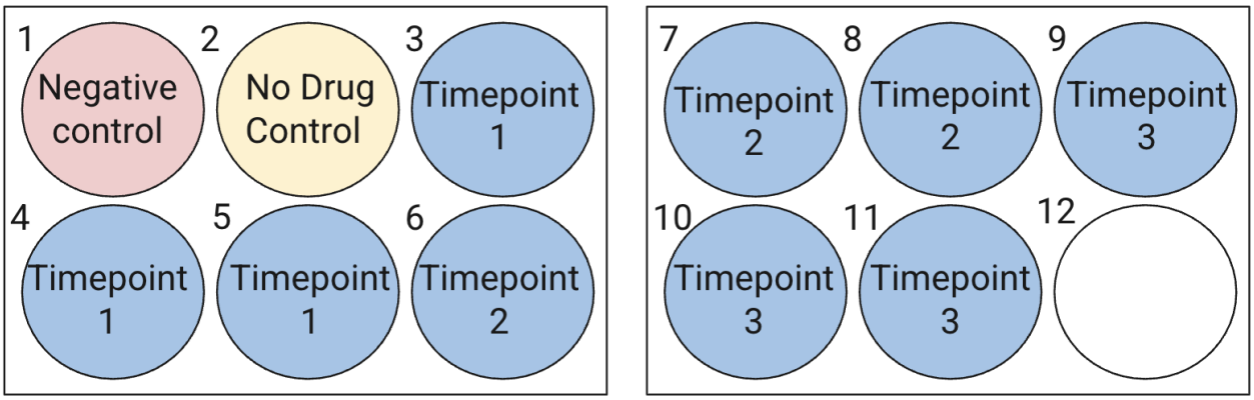

Figure 1: Layout of Control and Experimental Wells.

Two 6 well plates were used to culture the 3T3-L1 cells.

Wells 1-2: Control wells. The negative control (red) contained only CM at all stages of the procedure. The no drug control (yellow) was subject to normal adipogenic protocol; no drug was added.

Wells 3-5: First set of experimental wells that were subject to quercetin during induction of differentiation (first timepoint).

Wells 6-8: Second set of experimental wells that were subject to quercetin during adipocyte maturation (second timepoint).

Wells 9-11: Third set of experimental wells that were subject to quercetin during adipocyte maintenance (third timepoint)

2.2 Experimental procedure

The team started by plating and seeding the 3T3-L1 cells in each of the 11 wells. A 10 µl of 3T3-L1 cell suspension was pipetted into 2 ml of CM in each well. The plates were then incubated in 37°C for 1 week. With the essential nutrients and optimal temperature, the cells were confluent in 1 week.

A quercetin stock solution was created by dissolving 30 mg of the quercetin drug in 1 ml of DMSO to ensure proper control of quercetin concentration in each well. Before use, the stock solution was filtered using a syringe and filter to ensure no contamination was introduced into the wells.

After confirming confluency, the team conducted induction of differentiation. CM was removed with an aspirating device and replaced with DMI to provide the specific hormonal and chemical cues that trigger adipogenic differentiation. The first set of experimental wells (wells 3-5) received 1.8 ml of DMI and 200 µl of quercetin stock solution to achieve a 1:10 dilution of the drug. The negative control (well 1) received 2 ml of CM while the rest of the wells received 2 ml of DMI. This step in the procedure marks the first addition of quercetin in the experiment.

After six days, the team transitioned into the adipocyte maturation phase. The DMI was removed from wells 2-11 and CM was removed from well 1 using an aspirating device. The media removed was replaced with DMII, supporting the maturation process. Wells 6-8 received 1.8 ml of DMII and 200 µl of quercetin stock solution. The well 1 received 2 ml of CM, wells 9-11 received 1.8 ml of DMII as they received quercetin two days later, and the rest of the wells received 2 ml of DMII. At this stage, cells transition from the commitment phase to a stable, mature phenotype, making it an ideal time to examine how quercetin modulates downstream differentiation events. A couple days later, 200 µl of quercetin stock solution was added to wells 9-11. Adding quercetin during this later stage of the differentiation process helps assess whether the drug has any additional effects when the cells are committed to the adipocyte lineage.

Four days later, all media was aspirated from each well and replaced with 2 ml of CM, removing the strong differentiating signals. This enables the cells to settle into their mature adipocyte state without continual stimulation. This last step marked the end of the adipogenic protocol, enabling the analysis and quantification of lipid accumulation.

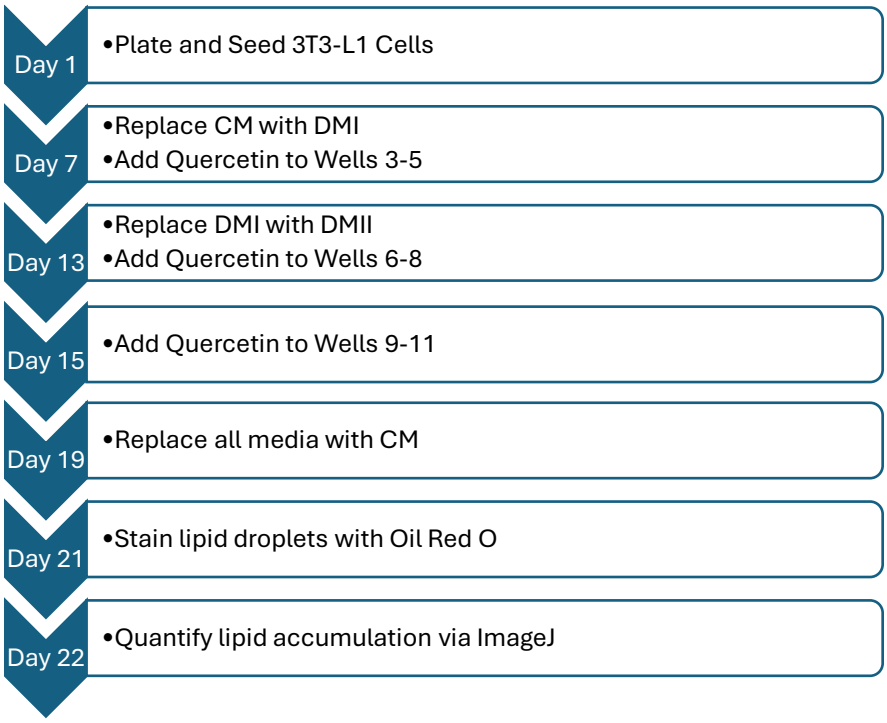

Figure 2: Timeline of Experimental Protocol
This protocol was based of a STAR protocol and altered[6].

2.3 Methods of Lipid Accumulation Analysis

During adipogenesis, the cells go through significant morphological changes as they differentiate from pre-adipocytes to mature adipocytes. One method the team used to analyze the lipid accumulation of the cells is through visualization under a microscope to directly observe morphological changes and understand quercetin's impact on cellular structure. Microscopic visualization enables comparative analysis between the control and experimental wells. Differences in cell morphology such as changes in cell size, number of lipid droplets, or changes in adipocyte structures may indicate the effects quercetin has.

Another method used to quantify the lipid accumulation was Oil Red O (ORO) staining. ORO is a dye that selectively binds to lipid droplets in fat cells, providing a quantitative assessment of lipid storage. Adipocyte differentiation leads to an increase in lipid droplets. Therefore, the amount and intensity of the ORO staining is a direct measure of lipid accumulation, serving as a reliable indicator for the extent to which differentiation has occurred. The procedure begins by removing any media in the wells with an aspirating device and replacing it with 1 ml of formalin. The cells are incubated at room temperature for five minutes. Then, the initial formalin is removed, and cells are washed with double distilled water. Another 1 ml of formalin is added to the wells and incubated for one hour at room temperature. The formalin is removed, and 1 ml of 60% isopropanol is added to each well. This step enhances the dye's ability to penetrate and bind to intracellular lipids. The isopropanol is removed, and 1 ml of ORO solution is added to each well, incubating for 10 minutes at room. Next, the ORO solution is aspirated out and the wells are washed five times with double distilled water to remove

unbound dye. Lastly, the cells are covered with 1 ml of double distilled water and stored in the fridge. At this stage, staining is complete, and the cells are ready for visualization with ImageJ, an online software used for lipid quantification.

ImageJ requires an upload of images of each well. The team took two pictures of each well at random positions within the well with a bright-field microscope. Each image was uploaded to ImageJ where the total area stained and total stain intensity for each well was quantified.

2.4 Key Resource Table

Culture Media (CM)	
Reagent	**Identification (lot #)**
DMEM	N/A
FBS	2993824
Penicillin-Streptomycin (Pen/Strep)	N/A
Differentiation Media I (DMI)	
Reagent	**Identification (lot #)**
Insulin	341615
Dexamethasone	BCCL5413
IBMX	STBF606IV
Rosiglitazone	392022
CM	2993824
Differentiation Media II (DMII)	
Reagent	**Identification (lot #)**
Insulin	341615
CM	2993824
Quercetin Stock Solution (100mM)	
Reagent	**Identification (lot #)**
Quercetin Solid Drug	0000431865
Dimethyl Sulfoxide (DMSO)	244325
Oil Red O Staining	
Reagent	
ORO Solution	
60% Isopropanol	
Formalin	
Double Distilled Water	
Other Equipment	
3T3-L1 cell line	
Pipettes	
Tissue Culture Hood	
Aspirating Device	
6-Well Plates	
Bright-Field Microscope	
Eppendorf Tubes	
Test Tubes	

Table 1: Layout of Key Reagents and Equipment for Experimental Protocol
Note: Proper sterilization techniques and personal protective equipment are needed to prevent contamination.

3.Results

ImageJ quantification of lipid accumulation revealed the following average areas stained for each experimental condition:

- **Well 1**: 1.09 px^2
- **Well 2**: 16.59 px^2
- **Wells 3-5**: 6.93 px^2
- **Wells 6-8**: 18.46 px^2
- **Wells 9-11**: 20.74 px^2

Table 2 shows the **average area stained** for each individual well.
The average integrative density of the lipid droplets for each well are as follows:

- **Well 1**: 268.66 px
- **Well 2**: 1125.13 px
- **Wells 3-5:** 334.32 px
- **Wells 6-8**: 988.42 px
- **Wells 9-11**: 1633.12 px

The average integrative density for each individual well is shown in **Table 2**.

Wells	**Area Stained (px^2)**	**Integrative Density (px)**
Well 1	1.09	268.66
Well 2	16.47	1125.13
Well 3	4.22	143.34
Well 4	7.22	255.35
Well 5	9.34	604.28
Well 6	19.06	687.79
Well 7	17.09	845.00
Well 8	19.22	1432.47
Well 9	16.49	766.08
Well 10	19.98	1849.67
Well 11	25.77	2283.61

Table 2: Summary of ImageJ Average Area Stained and Integrative Density Data for Each Well

Note: Two images were captured and quantified for each well. The averages of those two images were taken for each well. The data measured for area is in pixels squared and the data measure for integrative density is in pixels.

3.1 Quercetin's Effects on Lipid Accumulation are Dependent on the Experimental Conditions

The results ImageJ analysis results reveal that the negative control (Well 1) had both a low average area stained and low integrative density values, indicating minimal lipid accumulation. The no drug control (Well 2) had a moderate amount of lipid accumulation, shown by the relatively higher area stained and integrative density values. This no drug control is a baseline for comparison with each experimental conditions to understand the effects of quercetin in adipogenesis. When comparing the experimental wells to the baseline, the following observations were made:

- **First set of experimental wells (Well 3-5):** These wells showed lower area stained and integrative density compared to the baseline.
- **Second set of experimental wells (Well 6-8):** The area stained and integrative density in these wells were slightly higher than the baseline, but still modest compared to the third set.
- **Third set of experimental wells (Well 9-11):** In these wells, area stained and integrative density were significantly higher than in the baseline.

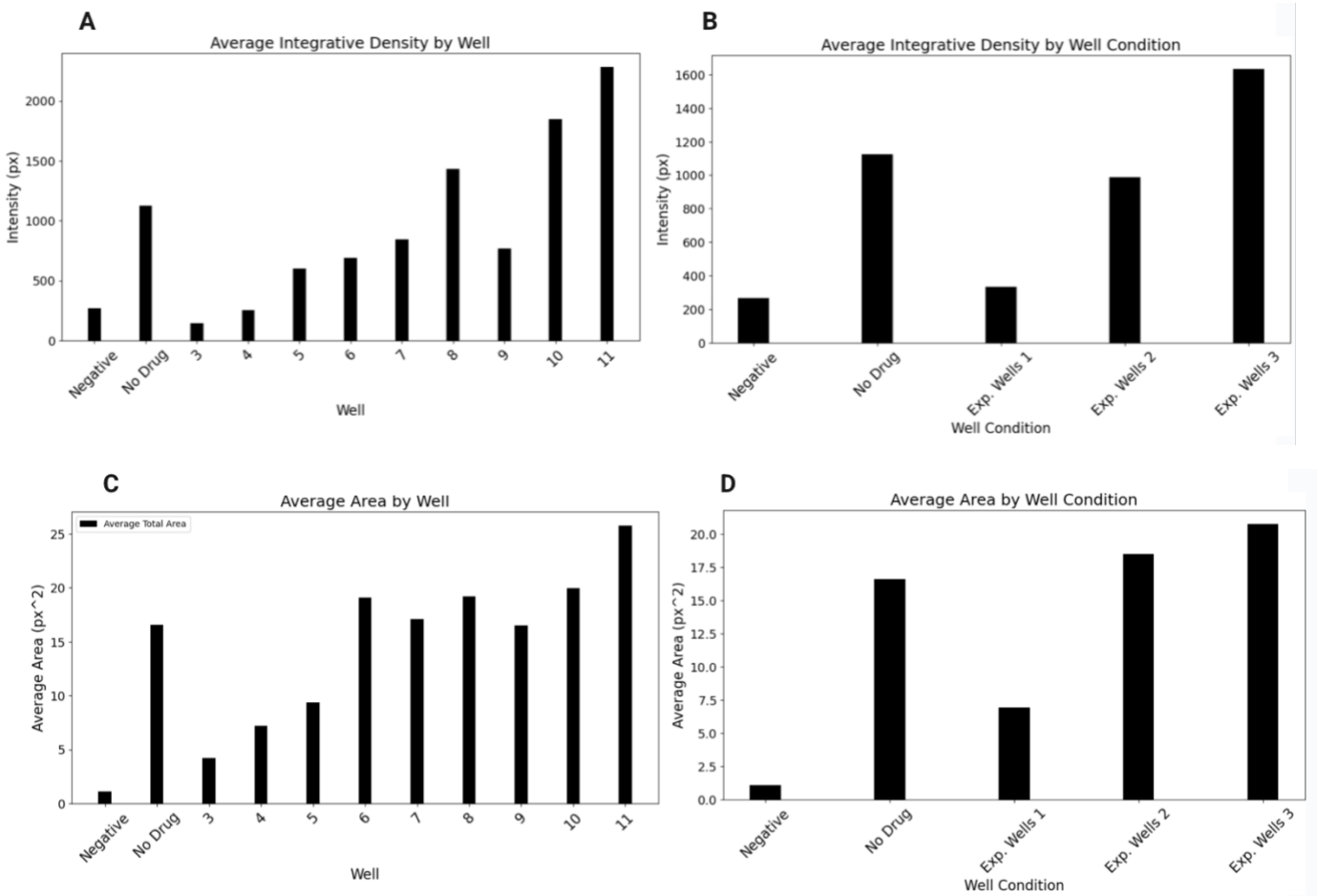

Figure 3: Bar graphs of ImageJ Quantification
A) Average integrative density for each well.
B) Average integrative density for each well condition. The average of all the wells in each set of experimental wells were calculated.
C) Average area stained for each well.
D) Average area stained for each well condition. The average of all the wells in each set of experimental wells were calculated.
Note: Two images were captured and quantified for each well. The averages of those two images were taken for each well. The data measured for area is in pixels squared and the data measure for integrative density is in pixels.

Overall, the effects of quercetin on lipid accumulation are condition dependent, differing depending on the conditions the cells are in at different stages of the adipogenic protocol.

3.2 Extended Exposure to Quercetin in 3T3-L1 cells Further Inhibits Adipogenesis

The timing of quercetin administration during the adipogenic protocol significantly influenced adipocyte differentiation. Wells 3-5 had the lowest lipid accumulation among all experimental conditions, indicating that early exposure to quercetin may inhibit adipogenesis the most, preventing fat cell formation. In contrast, wells 6-8

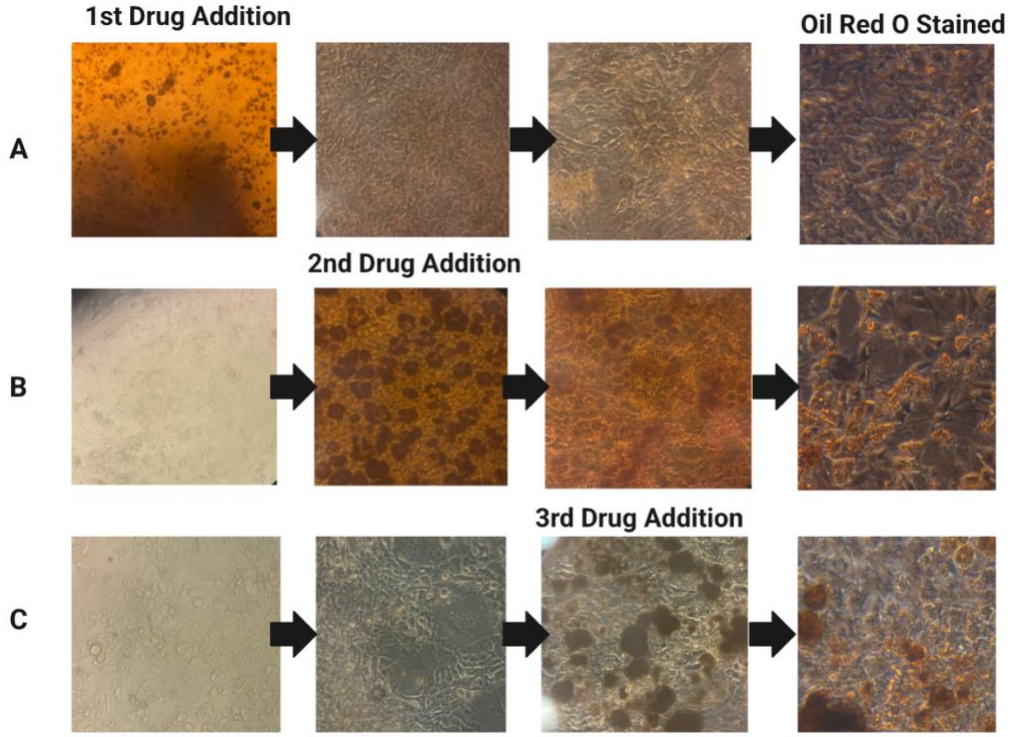

Figure 4: Visual Progression After Each Drug Addition During Adipogenesis
A) Progression of well 4, a well in the first set of experimental conditions. The cells go from being small, circular, and less confluent in early stages of adipogeneses to much more layered with an elongated shape in the later stages of adipogenesis.
B) Progression of well 6, a well in the second set of experimental conditions. The cells go from being

small and circular in the earliest stage, to larger and layered in the third stage with dark spots indicating infiltration of quercetin, to clustered and elongated in later stages of adipogenesis.
C) Progression of well 10, a well in the third set of experimental conditions. The cells go from being small and circular in the earliest stage, to more connected and layered with round droplets, to very layered with dark spots being the quercetin, to a variety of cell sizes with some cells seeming to be undergoing cell lysis.

showed moderate lipid accumulation, suggesting that quercetin's inhibitory effect on adipogenesis was reduced. Wells 9-11 displayed the highest lipid accumulation among the experimental conditions, suggesting that quercetin had the least inhibitory effect on adipogenesis in that experimental condition.

A progression of a well from each of the experimental wells throughout the adipogenic protocol is shown in **figure 4,** providing a more visual representation of how the timing of quercetin treatment impacts adipogenesis. In all experimental wells, the first exposure to quercetin is marked by the presence of dark spots. In the wells exposed to quercetin at the earliest stage of adipogenesis, the cells appear smaller, with reduced lipid droplets compared to the other wells. In contrast, the wells exposed to quercetin at the second earliest stage exhibit larger, more circular cells and larger lipid droplets. Finally, the wells exposed to quercetin at the latest stage show varying sizes with some cells seeming to undergo cell lysis.

3.3 Quercetin Effects the Morphology of Adipocytes

Quercetin significantly influenced the morphology of adipocytes, with the timing of its administration causing notable changes in the cellular structure and lipid accumulation. A visual representation of the cells after ORO staining, is shown in **Figure 5.**In well 1 the cells were highly confluent, forming a dense layer of 3T3-L1 cells. These cells appeared tightly aggregated and exhibited a slightly ovular shape, consistent with undifferentiated fibroblasts. In well 2, the cells exhibited a more rounded morphology with bright staining, which indicates normal adipocyte differentiation. These cells showed larger lipid droplets, confirming successful adipogenesis in the absence of treatment. Wells 3-5 showed smaller cells stacked on top of each other. These cells contained tiny lipid droplets, and extensive clustering was observed. In the wells 6-8, the cells exhibited a more elongated, fibroblast-like appearance, though some circular cells with lipid droplets were still present. These cells appeared strung together, with heterogeneous clustering in some areas. Finally, wells 9-11, showed signs of cellular lysis. The cells were more spread out and less clustered compared to those in the other wells. There was a mix of larger and smaller cells, indicating variability in differentiation.

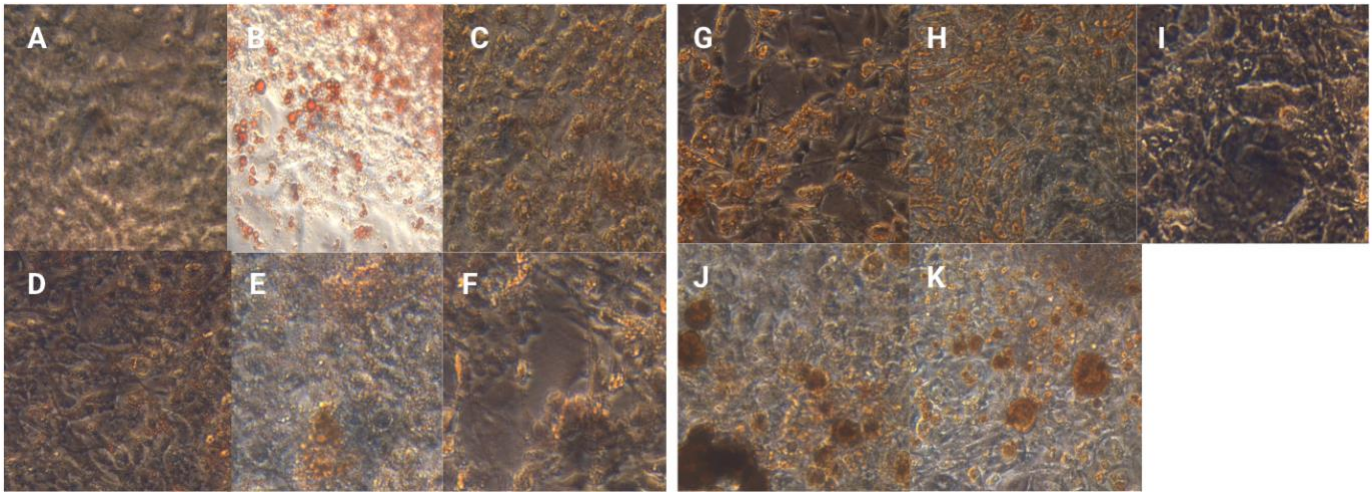

Figure 5: Effects of Quercetin on Cellular Morphology

A-K) Images of Wells 1-11 captured by a bright-field microscope after ORO staining. The contrast and brightness of the images were altered for better viewing of the ORO staining

Note: These images were very rushed due to time constraints which may lead to inaccurate results. Furthermore, quercetin's color could have inhibited the intensity of the ORO staining.

4.Discussion

4.1 Analysis of Results Relative to Hypotheses

Prior to the experimental protocol, the team proposed two hypotheses. The first hypothesis was that quercetin would counteract adipogenesis, predicting that lipid accumulation in wells 3-11 would be lower than in well 2. This hypothesis was based on previous studies showing that quercetin reduces lipid accumulation in HepG2 cells by inhibiting DNL, particularly through the inhibition of key transcription factors like SREBP-1 and XBP-1. The second hypothesis proposed that earlier quercetin exposure would have a greater inhibitory effect on adipogenesis compared to later treatments. The expectation was that wells 3-5 would show the least lipid accumulation, wells 6-8 would show moderate lipid accumulation, and wells 9-11 would have the most lipid accumulation, based on the understanding that adipogenesis is most vulnerable early in the differentiation process.

The experimental results did not fully support the first hypothesis. While the first set of experimental wells (well 3-5) showed reduced lipid accumulation relative to the baseline control, the second and third experimental wells (well 6-11) displayed unexpectedly higher lipid accumulation than the control, suggesting that quercetin had a minimal effect or potentially a stimulatory effect on adipogenesis during these later stages. However, the data comparing lipid accumulation between experimental wells supported the second hypothesis, with the first experimental wells showing the least lipid accumulation, the second experimental wells showing moderate lipid accumulation, and the third experimental wells showing the most.

In addition to lipid accumulation, cellular morphology observations revealed unexpected findings. While the anticipation was that all cells would adopt a circular adipocyte morphology by the end of adipogenesis, with varying lipid droplet sizes, some experimental wells exhibited fibroblast-like, elongated cells and signs of cell lysis. These morphological changes suggest that quercetin may have influenced cell behavior and differentiation in ways that were not entirely expected.

4.2 Possible Explanations for the Observed Outcomes

The first hypothesis, which predicted that quercetin would counteract adipogenesis in all experimental wells, was not supported by the results. It is important to recognize that the foundation of this hypothesis was based on an incomplete understanding. The experimental setup involving HepG2 liver cells and the 3T3-L1 adipocytes model contained several key differences, particularly in the lipid metabolism pathways of these two cell types. The liver, as a central organ in lipid metabolism, plays a crucial role in lipid synthesis and lipid export throughout the body. Quercetin's impact on lipogenic genes is likely more effective in liver cells because these pathways are central to lipid accumulation regulation, especially under conditions of insulin resistance and high free fatty acids. In contrast, adipocytes store lipids primarily through lipid droplet formation, a process regulated more by insulin and glucose signaling than by lipogenic transcription factors like SREBP-1 and XBP-1. This reliance on insulin and glucose signaling may make adipocytes less responsive to quercetin's effects on lipogenesis.

Furthermore, the experiment with HepG2 cells focused on immature, undifferentiated liver cells, which are likely more responsive to quercetin's lipid metabolism modulation. By contrast, 3T3-L1 adipocytes were exposed to quercetin at various stages of differentiation, leading to differing responses based on the stage of differentiation. The assumption that quercetin would behave similarly across different cell types and stages of differentiation was therefore uninformed. The discrepancy between the results of the liver and adipocyte experiments can be attributed to these differences in cell type and differentiation stage, which were not adequately accounted for in the initial hypothesis.

The second hypothesis was more supported by the results, suggesting that quercetin's effect on lipid accumulation is stage dependent. During the early stages of adipogenesis, pre-adipocytes are in the process of committing to the adipocyte lineage, involving the activation of specific transcription factors like PPARγ and C/EBPα. At this stage, the cells are highly sensitive to external factors, such as quercetin, making them more susceptible to external factors. At later stages of adipogenesis, the cells may have already fully committed to the adipocyte lineage, with the differentiation machinery already activated. As a result, the lipogenic genes involved in lipid synthesis may become resistant to further inhibition by quercetin, reducing the cells' responsiveness to external modulation. The observed increase in lipid accumulation in the third set of experimental wells could be explained by cellular stress responses to quercetin at later stages of adipogenesis. While quercetin's anti-inflammatory and antioxidant properties may have been beneficial in the early stages, it may have caused oxidative stress or ER stress in the later stages, leading to an increase in lipid accumulation due to cellular disruption.

Regarding the unexpected morphological responses, the observed changes seem to be stress related[7]. Early exposure to quercetin may have triggered cellular stress, leading to reduced cell size and smaller lipid droplets. However, in later stages, quercetin might have caused cellular damage, as seen with the lysis of cells in the third experimental wells, possibly due to oxidative stress or interference with lipid metabolism. The presence of Oil Red O staining in the lysing cells suggests that lipid droplets were still present in some of the damaged cells, indicating that quercetin may also be involved in modulating lipid storage processes under stress conditions.

4.3 Potential Errors and Limitations Impacting the Results

Due to timing constraints the team had to constantly adjust the experimental protocol while it was being conducted. While this flexibility enabled the completion of the project, the adjustments could have potentially impacted the results as the well-known adipogenic protocol was not followed perfectly. Furthermore, the team was very time crunched near the end, specifically when trying to obtain the images of the wells with the bright-field microscope for ImageJ analysis. Thus, the images might not have properly captured the intensity of the Oil Red O staining and the amount of staining. This affects the validity of our results. Therefore, the conclusions that are made are not concrete. More in-depth analysis and experimental replication are needed to make more concrete conclusions.

4.4 Future Research Directions

Much more work is required to understand the true effects of quercetin on adipogenesis. Given more time and funding, future experiments could involve more molecular assays to investigate the specific lipid metabolism pathways affected by quercetin. This could include quantitative PCR or western blotting to measure expression levels of key transcription factors. Furthermore, because ER stress and oxidative stress seem to play a role in the increase in lipid accumulation at later stages, future studies could explore quercetin's impact on stress pathways with flow cytometry assays to analyze markers for oxidative stress and ER stress. Lastly, because this

current study focuses on the short-term effects of quercetin, future studies could analyze long-term effects, specifically under conditions that mimic metabolic diseases like obesity or NAFLD.

5.Acknowledgements

We would like to thank the instructional team of BioE 115 for providing materials and guidance throughout this project. We want to thank the Bioengineering Department for funding, lab space, and equipment.

6.Author Contributions

A.G posed the idea to test quercetin on adipogenesis.
A.G, F.C, and V.R conducted research to develop and refine the experimental protocol and executed protocol steps.
F.C created the figures, analyzed the data, interpreted the results, and drafted the final paper.

7.Declaration of Interest

The authors declare no competing interests.

8. Declaration of Generative AI and AI-assisted Technologies

During the preparation of this work, the author used ChatGPT to navigate Word and formatting help. After using this tool or service, the author reviewed and edited the content as needed and full responsibility for the content of the publication.

7.References

1. Obesity. World Health Organization. Accessed May 10, 2025. https://www.who.int/health-topics/obesity#tab=tab_1.

2. Guerreiro VA, Carvalho D, Freitas P. Obesity, adipose tissue, and inflammation answered in questions. Journal of obesity. January 22, 2022. Accessed May 10, 2025. https://pmc.ncbi.nlm.nih.gov/articles/PMC8938152/.

3. Yang D, Wang T, Long M, Li P. Quercetin: Its main pharmacological activity and potential application in Clinical Medicine. Oxidative Medicine and Cellular Longevity. 2020;2020:1-13. doi:10.1155/2020/8825387

4. Wang Y, Li Z, He J, Zhao Y. Quercetin regulates lipid metabolism and fat accumulation by regulating inflammatory responses and glycometabolism pathways: A Review. Nutrients. 2024;16(8):1102. doi:10.3390/nu16081102

5. Gnoni A, Di Chiara Stanca B, Giannotti L, Gnoni GV, Siculella L, Damiano F. Quercetin reduces lipid accumulation in a cell model of NAFLD by inhibiting de novo fatty acid

synthesis through the acetyl-COA carboxylase 1/AMPK/PP2A axis. International Journal of Molecular Sciences. 2022;23(3):1044. doi:10.3390/ijms23031044

6. Strnadová M, Thor D, Kaczmarek I. Protocol for changing gene expression in 3T3-L1 (pre)adipocytes using Sirna-mediated knockdown. STAR Protocols. 2024;5(2):103075. doi:10.1016/j.xpro.2024.103075

7. Xue M, Liao Y, Jiang W. Insights into the molecular changes of adipocyte dedifferentiation and its future research opportunities. Journal of Lipid Research. September 18, 2024. Accessed May 10, 2025. https://www.sciencedirect.com/science/article/pii/S0022227524001494.